MedCare Hospital Readmission Reduction & Patient Care Optimization Project

Executive Findings & Strategic Recommendations Report

Prepared For: MedCare Hospital Executive Leadership Team
Prepared By: Matthew Nduva, Analytics Consultant
Project Period Analyzed: January 2023 – December 2025
Dataset Size: 5,000 Patient Records

Executive Summary

MedCare Hospital commissioned this analysis to understand the drivers of patient readmissions, identify opportunities to improve patient outcomes, optimize healthcare delivery, and reduce unnecessary costs associated with repeat hospital visits.

The project combined Business Intelligence (SQL Analytics), Predictive Machine Learning, and Executive Dashboarding to uncover actionable insights across hospital operations.

Our analysis revealed that patient readmissions are significantly influenced by chronic conditions, incomplete follow-up care, previous admission history, patient satisfaction levels, and length of hospital stay.

Using machine learning, we successfully developed a predictive model capable of identifying high-risk patients with approximately 75% accuracy and an 81% readmission detection rate, allowing the hospital to proactively intervene before costly readmissions occur.

Business Questions Addressed

The project was designed to answer the following executive questions:

Operational Performance

- What is the overall cost of patient treatment?

- Which departments consume the most resources?
- Which departments experience the highest readmission rates?
- How long are patients staying in the hospital?

Patient Outcome Analysis

- Which patient groups are most likely to be readmitted?
- Which diagnoses contribute most to readmissions?
- Does patient satisfaction impact readmission likelihood?
- Does follow-up care reduce readmissions?

Financial Performance

- What is the financial impact of readmitted patients?
- Which departments generate the highest treatment costs?
- Which patient segments create the highest readmission burden?

Predictive Intelligence

- Can future readmissions be predicted before discharge?
- Which factors are the strongest indicators of readmission risk?
- How can hospital leadership reduce avoidable readmissions?

Section 1: Operational & Financial Performance Analysis

Overall Hospital Performance

KPI	Value
Total Treatment Cost	$116.91 Million
Average Treatment Cost	$23,381
Average Length of Stay	8 Days
Average Satisfaction Score	5.43 / 10
Total Patients Analyzed	5,000

Executive Insight

The hospital managed a substantial patient volume while incurring over $116 million in treatment costs during the analysis period. However, the average patient satisfaction score of 5.43 indicates significant room for improvement in patient experience and care quality.

Department Performance Analysis

Highest Patient Volumes

Department	Patients
Orthopedics	644
General Medicine	643
Cardiology	636
Pulmonology	625
Oncology	622

Executive Insight

Patient demand is relatively balanced across departments, indicating consistent utilization of hospital services.

Highest Treatment Costs

Department	Total Cost
Orthopedics	$15.59M
General Medicine	$15.30M
Cardiology	$14.76M
Oncology	$14.73M
Pulmonology	$14.49M

Executive Insight

Orthopedics and General Medicine consume the highest share of hospital resources and should remain priority areas for operational efficiency reviews.

Average Length of Stay by Department

Department	Avg Stay
General Medicine	8.42 Days
Oncology	8.22 Days
Emergency	8.03 Days
Pulmonology	8.00 Days

Executive Insight

Departments with longer patient stays tend to have more complex cases and higher treatment costs. Extended stays also increase the probability of future readmissions.

Section 2: Readmission Analysis

Department Readmission Rates

Department	Readmission Rate
Oncology	44.05%
Emergency	41.76%
General Medicine	41.21%
Cardiology	40.72%
Orthopedics	40.22%

Executive Finding

Oncology patients experience the highest readmission rate across the hospital.

This suggests a need for stronger post-discharge monitoring, care coordination, and patient education programs within oncology services.

Financial Impact of Readmissions

Total Cost Associated with Readmitted Patients

$47.08 Million

Executive Finding

Approximately 40% of total treatment spending is associated with patients who return for additional care.

Reducing readmissions by even 10–15% could generate millions in annual cost savings.

Age Group Risk Analysis

Age Group	Readmission Rate
60+ Years	49.10%
31–45 Years	34.01%
46–60 Years	33.89%
18–30 Years	33.07%

Executive Finding

Patients aged 60 and above are significantly more likely to return to the hospital.

This demographic should be prioritized for preventive care initiatives and follow-up programs.

Diagnosis Analysis

Conditions Most Associated with Readmissions

Diagnosis	Readmissions
Asthma	193

Hypertension	175
Infection	160
Breast Cancer	99
Diabetes	93

Executive Finding

Most readmissions are linked to chronic and long-term health conditions, reinforcing the need for continuous patient management programs.

Patient Satisfaction Analysis

Satisfaction Level	Readmission Rate
Low Satisfaction	52.30%
Medium Satisfaction	30.70%
High Satisfaction	32.31%

Executive Finding

Patients reporting low satisfaction are substantially more likely to return to the hospital.

Improving patient experience may directly contribute to reducing readmission rates.

Insurance Analysis

Insurance Type	Readmission Rate
Government	40.61%
Self-Pay	40.39%
Employer	40.15%
Private	39.65%

Executive Finding

Insurance type has minimal impact on readmission risk, suggesting clinical and behavioral factors are more influential than payment method.

Readmission Trends over Time

Monthly readmission activity remained consistently high across all three years analyzed, indicating that readmissions are a persistent operational challenge rather than a temporary issue.

No significant downward trend was observed.

Section 3: Predictive Machine Learning Findings

Objective

Develop a predictive model capable of identifying patients likely to be readmitted before discharge.

Final Model Performance

XGBoost Classifier

Metric	Result
Accuracy	74.70%
Precision	65%
Recall	81%
F1 Score	72%

Executive Interpretation

The model successfully identifies 81% of future readmissions, making it a valuable decision-support tool for care teams.

The emphasis on recall is appropriate because missing a high-risk patient can result in significant clinical and financial consequences.

Key Drivers of Readmission

Rank	Driver	Importance
1	Chronic Condition	15.3%
2	Follow-Up Completed	14.4%
3	Previous Admissions	11.3%
4	Satisfaction Score	9.9%
5	Length of Stay	8.8%
6	Age	6.7%

What the Model Revealed

Chronic Conditions Drive Future Readmissions

Patients with chronic illnesses are the most likely to return after discharge.

Follow-Up Care Significantly Reduces Risk

Failure to complete follow-up appointments is one of the strongest predictors of readmission.

Past Admissions Predict Future Admissions

Patients who have previously been hospitalized multiple times are more likely to require future care.

Patient Experience Matters

Lower satisfaction scores correlate strongly with increased readmission risk.

Longer Hospital Stays Signal Higher Risk

Patients requiring extended treatment often need additional monitoring after discharge.

Strategic Recommendations

Recommendation 1: Launch a High-Risk Patient Monitoring Program

Use predictive analytics to identify high-risk patients before discharge and automatically enroll them in enhanced care programs.

Recommendation 2: Improve Follow-Up Completion Rates

Implement:

- Automated appointment scheduling
- SMS reminders
- Follow-up call programs
- Nurse outreach initiatives

Recommendation 3: Strengthen Chronic Disease Management

Focus additional resources on:

- Oncology
- Cardiology
- Pulmonology
- Diabetes Care
- Hypertension Management

Recommendation 4: Prioritize Senior Patient Care

Patients aged 60+ should receive:

- Enhanced discharge planning
- Medication reviews
- Caregiver education
- Home care support

Recommendation 5: Improve Patient Satisfaction

Target improvements in:

- Communication
- Wait times
- Discharge education
- Care coordination

Improving satisfaction scores is likely to reduce future readmissions.

Expected Business Impact

If MedCare Hospital implements the recommendations outlined in this report, leadership can expect:

- Reduced readmission rates
- Improved patient outcomes
- Lower operational costs
- Better bed utilization
- Increased patient satisfaction
- Enhanced quality-of-care performance metrics
- Data-driven clinical decision-making

Final Consultant Conclusion

The analysis confirms that readmissions are not random events. They can be anticipated using available patient, treatment, and operational data. By combining predictive analytics with targeted patient interventions, MedCare Hospital has a significant opportunity to improve patient outcomes while reducing millions of dollars in avoidable healthcare costs.

Prepared by:
Matthew Nduva

Analytics Consultant

DataVibe Analytics